

SULIT
SM015
Mathematics 1
Semester 1
Session 2024/2025

SM015
Matematik 1
Semester 1
Sesi 2024/2025



KEMENTERIAN PENDIDIKAN MALAYSIA

KOLEJ MATRIKULASI KEJURUTERAAN KEDAH

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION
PRA PSPM 1

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

Kertas soalan ini mengandungi **8** halaman bercetak.
This question paper consists 8 printed pages.

INSTRUCTIONS TO CANDIDATE:

This question paper consists of **Section A** and **Section B**.

Section A consists of 3 questions. **Section B** consists of 7 questions.

Answer all questions.

Marks for each question or section are shown in the bracket at the end of the question or section

All steps must be shown clearly.

Only non-programmable scientific calculator can be used.

Numerical answer may be given in the form of π , e , surd, simplest fractions or up to three significant figures, where appropriate, unless stated otherwise in the question.

ARAHAN KEPADA CALON

Kertas soalan ini mengandungi **Bahagian A** dan **Bahagian B**

Bahagian A mengandungi 3 soalan. **Bahagian B** mengandungi 7 soalan.

Jawab semua soalan.

Markah yang diperuntukan bagi setiap soalan atau bahagian soalan ditunjukkan dalam kurungan pada penghujung soalan atau bahagian soalan.

Semua langkah kerja hendaklah ditunjukkan dengan jelas.

Kalkulator saintifik yang tidak boleh diprogramkan sahaja yang boleh digunakan.

Jawapan berangka boleh diberi dalam bentuk π , e , surd, pecahna termudah atau sehingga tiga angka bererti, dimana-mana yang sesuai, kecuali jika dinyatakan dalam soalan.

LIST OF MATHEMATICAL FORMULAE

Differentiation

$f(x)$	$f'(x)$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$
$\cot x$	$-\operatorname{cosec}^2 x$
$\sec x$	$\sec x \tan x$
$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$

The Sum Rule

$$\frac{d}{dx}(u(x) + v(x)) = \frac{du}{dx} + \frac{dv}{dx}$$

Product Rule

$$\frac{d}{dx}(u(x) \cdot v(x)) = v \frac{du}{dx} + u \frac{dv}{dx}$$

Quotient Rule

$$\frac{d}{dx}\left(\frac{u(x)}{v(x)}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Chain Rule

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$$

Shape**Volume****Surface Area****Sphere**

$$V = \frac{4}{3} \pi r^3$$

$$S = 4\pi r^2$$

Right circular cone

$$V = \frac{1}{3} \pi r^2 h$$

$$S = \pi r^2 + \pi r l$$

Right circular cylinder

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$

Trigonometry:

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\sin A - \sin B = 2 \cos \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$$

$$\cos A - \cos B = -2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A$$

$$= 2 \cos^2 A - 1$$

$$= 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

SECTION A [25 marks]

This section consists of 3 questions. Answer **all** questions.

- 1 (a) Evaluate $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 3x}}{2x - 3}$ [4 marks]

(b) The function $f(x)$ defined as

$$f(x) = \begin{cases} px + 5, & x < 3 \\ q, & x = 3 \\ x^2 + 2x - 1, & x > 3 \end{cases}$$

Find values of p if $\lim_{x \rightarrow 3} f(x)$ exist. Hence, find the value of q such that

$f(x)$ is continuous at $x = 3$. [5 marks]

- 2 (a) If $5y^2 + 3y \sin x = 45$, find $\frac{dy}{dx}$. Hence evaluate $\frac{dy}{dx}$ when $x = 0$. [4 marks]

(b) Given that $f(x) = \frac{5}{x^2} + x$. By using the first principle, find $f'(x) = \frac{5}{x^2} + x$. [5 marks]

- 3 Given that $f(x) = x(x-2)^3$.

(a) Use the First Derivative Test to determine the relative extremum of the function. [4 marks]

(b) Find the point of inflection. [3 marks]

SECTION B [75 marks]

This section consists of 7 questions. Answer **all** questions.

- 1 Given that $z = 2\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$ and $\bar{z}$ its conjugate. Express $2\bar{z} - \frac{1}{z}$ in form of $a + bi$ where a and b are real numbers. Hence express $2\bar{z} - \frac{1}{z}$ in polar form. [5 marks]

- 2 (a) Solve $\log(x^2 + 5) - 1 = \log(x - 2)$. [3 marks]

- (b) Solve $\frac{2x+1}{x+3} \leq -1$ [4 marks]

- (c) Solve $|x+5| - 8 > 0$. [4 marks]

- 3 Consider functions of $f(x) = (x-1)^2 + 1$, $x < 1$

- (a) Find $f^{-1}(x)$ and its domain. [4 marks]

- (b) Sketch graph of $f(x)$ and $f^{-1}(x)$ on the same axes. [3 marks]

4. Given that $(g \circ f)(x) = 5 - x$ and $g(x) = \log_3(x - 2)$.

- (a) Determine $f(x)$. [3 marks]

- (b) Show that $f(x)$ is a one-to-one function by using algebraic approach. [2 marks]

- (c) Find $f^{-1}(x)$. State the domain and range of $f^{-1}(x)$ [5 marks]

- (d) If $h(x) = 5 - \log_3 \frac{1}{x+2}$, obtain $(f \circ h)(x)$. [3 marks]

5. The function f and g are defined respectively by

$$f(x) = 4 - 2x$$

$$g(x) = \frac{1-x}{3x+1}, x \neq -\frac{1}{3}$$

- (a) Find the function $T(x) = (f \circ g)(x)$ and state its domain [3 marks]
- (b) Determine the vertical and horizontal asymptotes of $T(x)$ [4 marks]
- (c) Sketch the graph of $y = T(x)$, clearly showing the asymptotes and axes intercepts [4 marks]
- (d) From part (c), state the range of $T(x)$ and show that $T(x)$ is a one-to-one function [3 marks]

- 6 Given $x = 2e^t$ and $y = e^{2t} - 1$.

- (a) Find the point (x, y) on the curve at which the gradient is 1. [11 marks]
- (b) Find the value of t so that

$$8\left(\frac{d^2y}{dx^2}\right) + 5\frac{dy}{dx} - 7 = 0$$
 [5 marks]

- 7 An open cylindrical wastepaper bin, of radius r cm and capacity V cm³, is to have a surface area of 5000cm^2

- (a) Show that $V = \frac{1}{2}r(5000 - \pi r^2)$. [4 marks]
- (b) Calculate the maximum possible capacity of the bin. [5 marks]

END OF QUESTION PAPER